Nightpass

Unlinkable prepaid API credits for AI agents — on Midnight

Pay for every API call with a zero-knowledge proof.
The provider learns "paid by a valid, unspent credit" — never *who*.

Midnight Buildathon · Wave 3 · Track: Build Privacy-First Apps

The problem

- AI agents now pay per API call (x402 micropayments, API keys tied to wallets).
- Every paid call is linkable: which agent, which endpoint, how often, when.
- For an agent, that log *is* its strategy: trading signals, research targets, customers.
- Today the only options are: trust the provider, or don't use paid APIs.

The idea

- **Buy once:** the agent buys a bundle of *N* calls; the chain stores only a hiding commitment `H(secret, N)`.
- **Spend anonymously:** each call = one ZK `redeem()` proving "I own an unspent use of *some* bundle in the tree".
- **No double spend:** a per-use nullifier `H(secret, i)` goes into a public set.
- **Bound to the request:** the API's nonce is recorded next to the nullifier → the server serves only paid requests.

Dual-ledger design

	Public ledger (on-chain)	Private state (agent only)
Bundle	commitment in `HistoricMerkleTree<16>`	secret, N, next index
Spend	nullifier in `Set`, nullifier → request tag in `Map`	which bundle, which use, Merkle path
Counters	`bundlesIssued`, `creditsRedeemed`	credits left
Operator	hash of the operator key (sealed)	operator secret

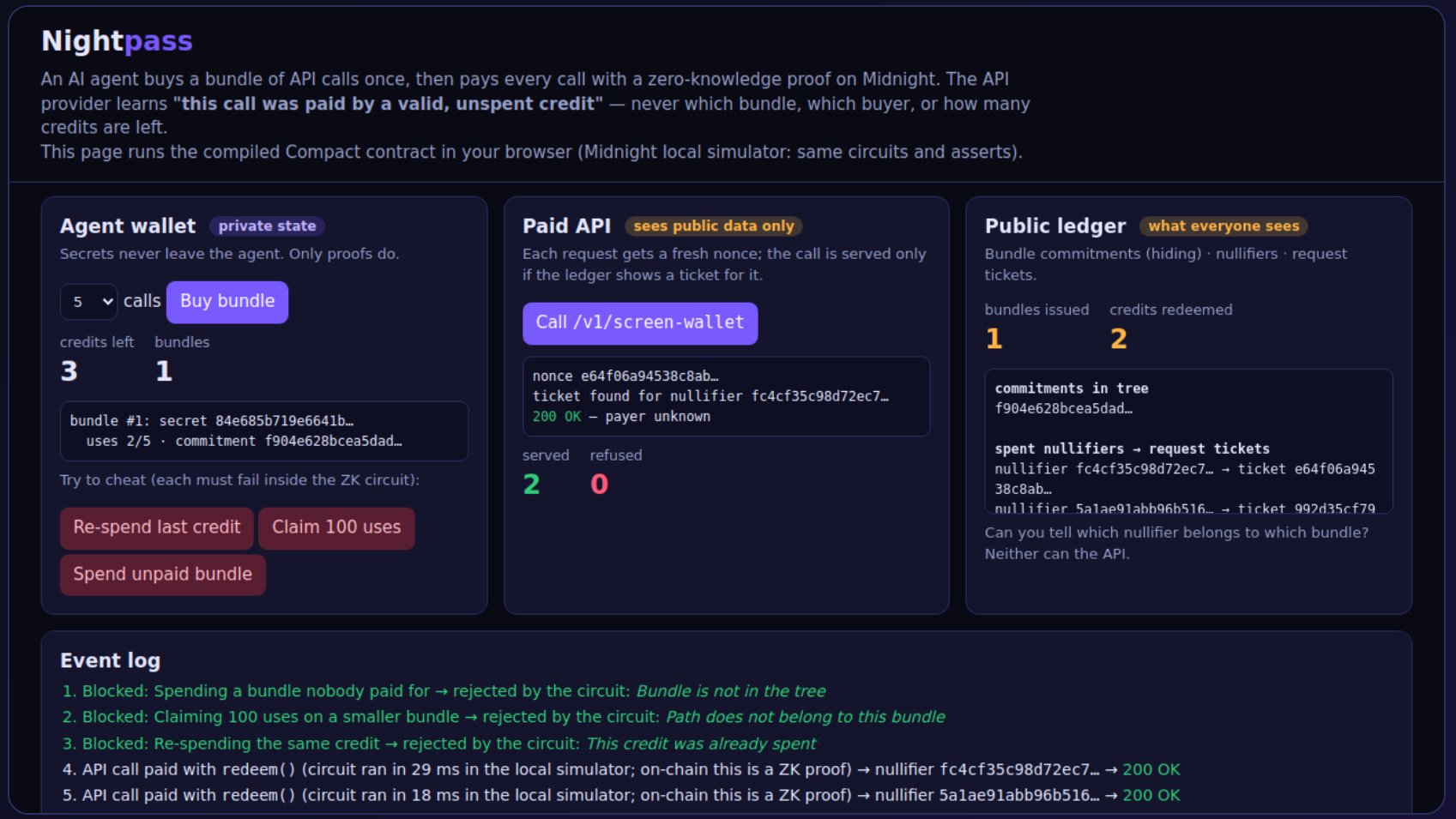
Witnesses: `bundle()`, `bundlePath()`, `useIndex()`, `localSecret()`. Historic roots keep old bundles spendable.

What the circuit enforces

- ✔ path leaf = `bundleCommitment(bundle)` → N cannot be inflated
- ✔ root is a current or past root of the tree → only paid bundles
- ✔ `useIndex < N` → exactly N spends
- ✔ nullifier unspent → no replay / double spend
- ✔ only the operator can `issue()`

10 simulator tests cover each rule plus unlinkability and request binding — all passing. Contract compiles with real ZK keys.

Demo



Live: nightpass-demo.nightshift-labs.workers.dev — the compiled contract running in the browser (local Midnight simulator), with three "try to cheat" buttons.

Market & adoption

- **Who pays:** API providers selling to agents (data, search, on-chain analytics) who want privacy-conscious agent customers.
- **Why now:** agent-to-API payments (x402) are growing; privacy is the missing piece.
- **Adoption path:** drop-in `X-Nightpass` header next to x402; we run our own paid agent API and will accept Nightpass first.
- **Model:** providers keep 100% of the bundle price; optional hosted verifier.

Roadmap & honest limits

- Next: Midnight testnet deploy + Lace wallet in the UI.
- Shielded payment inside the purchase, so buying is private too.
- x402 integration: verify the nullifier via the indexer in the API middleware.
- Bundle expiry, per-endpoint credit classes.
- Limits: the operator knows who bought which commitment; unlinkability starts after purchase. Anonymity set = bundles issued before the proven root.